

Table D.2 ORM field types

Field	Description
<pre>FileField(upload_to="", storage=None, max_length=100, **options)</pre>	This stores the path to a file that was uploaded. The <code>upload_to</code> argument indicates the directory for uploading. The <code>storage</code> argument indicates the object that handles file storage; if <code>None</code>, the default Django storage mechanism is used. Full details on storage objects are available in the documentation: https://docs.djangoproject.com/en/dev/topics/files/#storage-objects. The <code>max_length</code> argument restricts the length of the filename. Instances of this field have attributes that contain more information about the file: <code>.name</code> is the file name, <code>.path</code> is the file path, <code>.size</code> is the file size, and <code>.url</code> is the URL used to access the file. Field instances also support several methods—<code>.open()</code>, <code>.close()</code>, <code>.save()</code>, and <code>.delete()</code>—which correspond to the equivalent file operations.
<pre>FilePathField(path="", match=None, recursive=False, allow_files=True, allow_folders=False, max_length=100, **options)</pre>	This is a <code>CharField</code>, whose choices are restricted to the names of files at a given path. The <code>path</code> argument is required and points to the directory used to inspect for allowable values. All other arguments are optional. The <code>match</code> argument can contain a regular expression to restrict the names of the paths. Set <code>recursive</code> to <code>True</code> to include subdirectories. Values for <code>allow_files</code> and <code>allow_folders</code> indicate whether file and directory paths are included as possible values.
<pre>FloatField(**options)</pre>	Stores a floating point number.
<pre>GenericIPAddressField(protocol= "both", unpack_ipv4=False, **options)</pre>	Stores an IPv4 or IPv6 IP address as a string. The <code>protocol</code> argument can be one of <code>both</code>, <code>IPv4</code>, or <code>IPv6</code>. Setting <code>protocol</code> to <code>IPv4</code> or <code>IPv6</code> restrict the field to that kind of address. When <code>protocol</code> is <code>both</code> and <code>unpack_ipv4</code> is <code>True</code>, packed IP4 addresses are unpacked when accessed (e.g., <code>::ffff:192.0.2.1</code> becomes <code>192.0.2.1</code>).
<pre>ImageField(upload_to=None, height_field= None, width_field=None, max_length=100, **options)</pre>	An instance of a <code>FileField</code> that validates the uploaded file is an image. It requires the Pillow library to be installed. The <code>height_field</code> and <code>width_field</code> arguments optionally name other fields in the table that get automatically updated with the image's height and width values.